

**SCHOOL OF  
COMPUTING**

**Amrita Vishwa Vidyapeetham, Chennai Campus**  
**Department of Computer Science and Engineering**  
**Exercise-4: SVM**

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**CLASS : CSE-B**

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The SVM exercise asks you to generate a random dataset following a quadratic distribution and classify it using SVM.

```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

np.random.seed(42)

n = 500

x1 = np.random.uniform(-5, 5, n)
x2 = np.random.uniform(-2, 25, n)

y = (x2 > x1**2).astype(int)

X = np.column_stack((x1, x2))

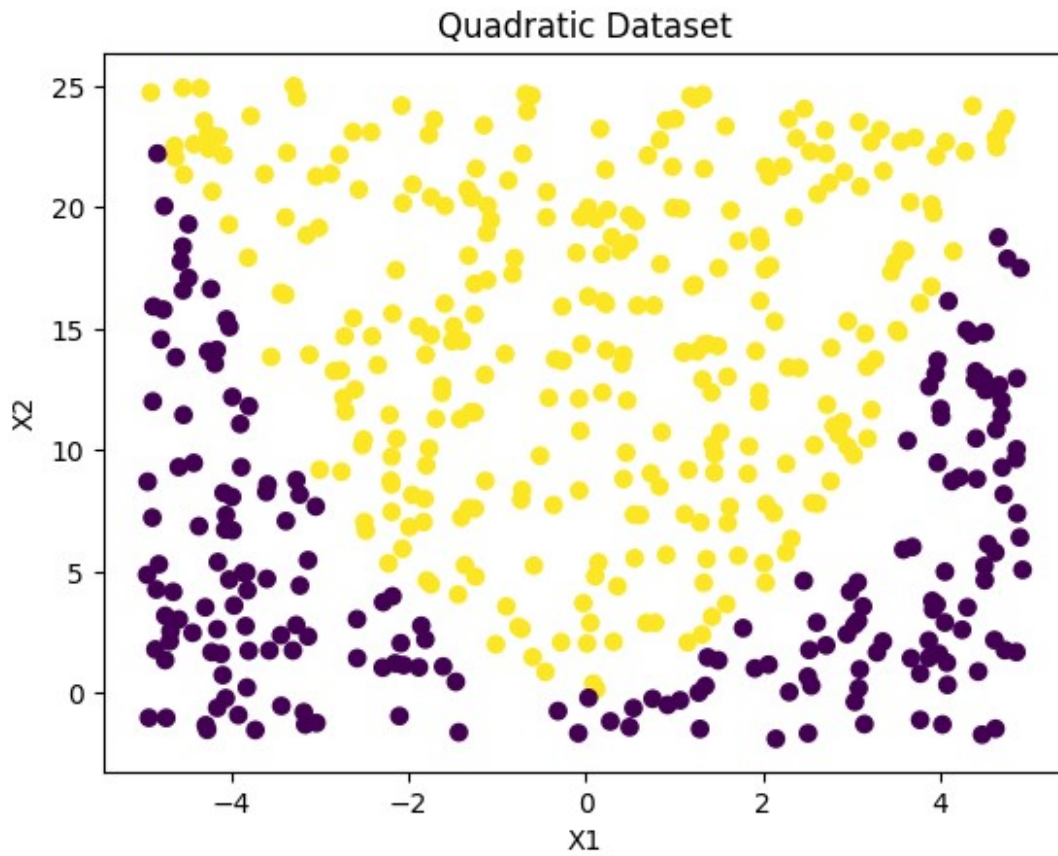
print("Dataset shape:", X.shape)
print("Target shape:", y.shape)

Dataset shape: (500, 2)
Target shape: (500,)

plt.scatter(
    X[:, 0],
    X[:, 1],
    c=y
)

plt.xlabel("X1")
plt.ylabel("X2")
plt.title("Quadratic Dataset")

plt.show()
```



```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y,  
    test_size=0.2,  
    random_state=42  
)
```

```
print("Training samples:", len(X_train))  
print("Testing samples:", len(X_test))
```

```
Training samples: 400
```

```
Testing samples: 100
```

```
model = SVC(  
    kernel='rbf',  
    random_state=42  
)
```

```
model.fit(X_train, y_train)
```

```
SVC(random_state=42)
```

```
y_pred = model.predict(X_test)
```

```

print("Predictions:")
print(y_pred)

Predictions:
[1 1 0 0 0 1 0 1 1 1 1 0 1 1 0 0 0 1 1 0 1 1 1 0 0 1 0 1 1 0 1 1
 0 0
 1 1 0 1 1 1 1 0 1 1 0 1 1 0 0 1 1 1 0 1 1 0 1 1 0 1 0 0 0 1 0 1 1 0 1
 1 0
 1 1 1 0 1 1 0 1 0 0 0 0 1 1 0 1 1 1 1 0 1 1 0 1 0 0]

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

Accuracy: 0.93
Accuracy Percentage: 93.0 %

x_min, x_max = X[:, 0].min() - 1, X[:, 0].max() + 1
y_min, y_max = X[:, 1].min() - 1, X[:, 1].max() + 1

xx, yy = np.meshgrid(
    np.linspace(x_min, x_max, 300),
    np.linspace(y_min, y_max, 300)
)

grid = np.c_[xx.ravel(), yy.ravel()]

Z = model.predict(grid)

Z = Z.reshape(xx.shape)

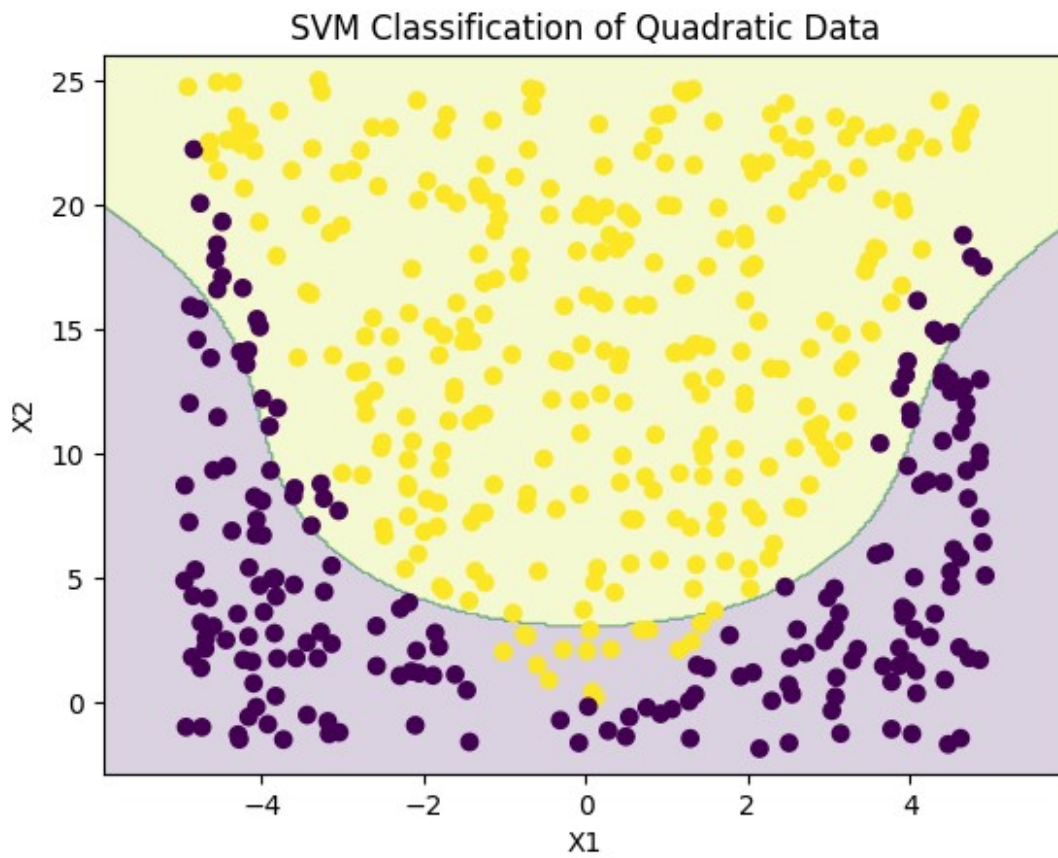
plt.contourf(xx, yy, Z, alpha=0.2)

plt.scatter(
    X[:, 0],
    X[:, 1],
    c=y
)

plt.xlabel("X1")
plt.ylabel("X2")
plt.title("SVM Classification of Quadratic Data")

plt.show()

```



Conclusion: An SVM classifier with an RBF kernel was successfully applied to a quadratic-distribution dataset. The model was trained, tested and evaluated using classification accuracy.